

YEAR 7 MATHEMATICS

Chapter 10

Analysing data

MIXED PRACTICE WORKSHEETS

10

How to use this booklet

- Each lesson is **one double-sided sheet**. Write your name, class and date at the top.
- Then work down the **columns**, left to right. They get harder as you go: **Easy** **Easy** **Medium** on page 1, then **Medium** **Challenging** **Challenging** on page 2.
- Most questions practise **today's skill**. A question marked **from 1.03** comes from an earlier lesson, to keep it fresh. Stuck? Look back at that lesson.
- Each question tells you what to do: read it carefully. Show your working on the lines. The first question with working in each Easy column shows you how to set it out.

WE DO

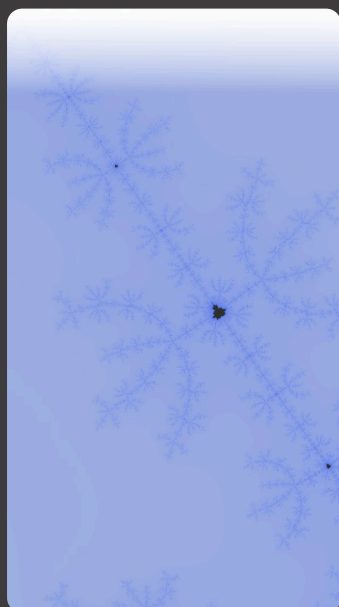
The grey box at the top. Your teacher shows the key idea with two or three examples. **Copy every step** into the space.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.01	Interpreting graphs	3		☐
10.02	Misleading graphs	5		☐
10.03	Dot plots	7		☐
10.04	Stem-and-leaf plots	9		☐
10.05	Mean and mode	11		☐
10.06	Median and range	13		☐
10.07	Analysing dot plots and stem-and-leaf plots	15		☐
10.08	Comparing data	17		☐

In this chapter you will

- interpret picture, column, sector and line graphs
- identify misleading graphs
- read and draw dot plots and stem-and-leaf plots
- find the mean, mode, median and range
- analyse data shown in plots
- compare two sets of data



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \rightarrow z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is snowflake-like branches between the main body and the top bulb. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

NAME _____ CLASS _____ DATE _____

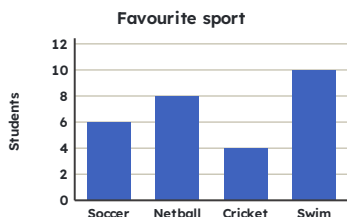
WE DO

KEY IDEA

Read the title, the axis labels and the scale first. Then read the value off the graph. A sector graph shows parts of a whole: the full circle is 100%.

Watch your teacher.
Copy every step.

E1 How many students chose Netball?



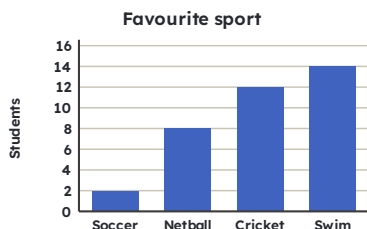
E2 How many more chose Cricket than Soccer?

E3 What fraction of the students chose Soccer?

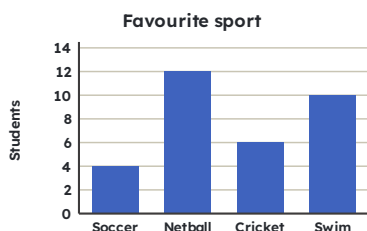
Easy

Questions 1–3

1 How many students chose Soccer?

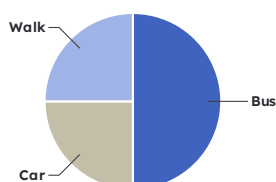


2 Which sport was the most popular?



3 What percentage of the class chose Walk?

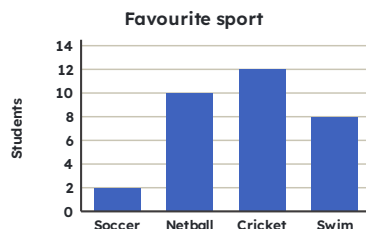
How students get to school



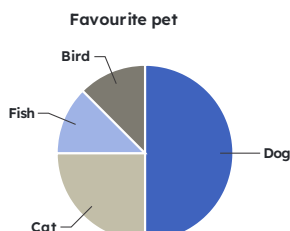
Easy

Questions 4–6

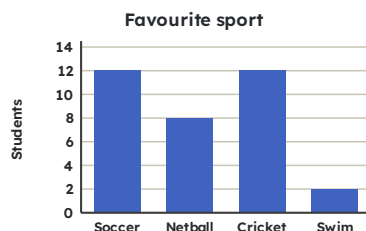
4 Which sport was the most popular?



5 What percentage of the class chose Bird?



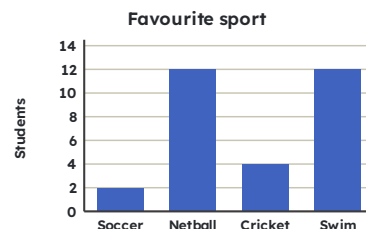
6 How many students chose Netball?



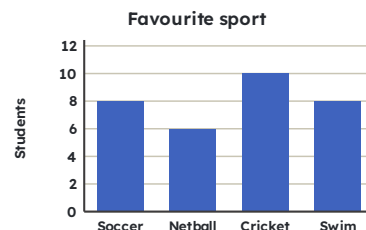
Medium

Questions 7–9

7 How many more students chose Cricket than Soccer?



8 How many students were surveyed altogether?



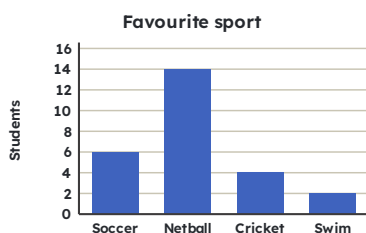
9 Write 3 out of 20 as a percentage.

from 4.12

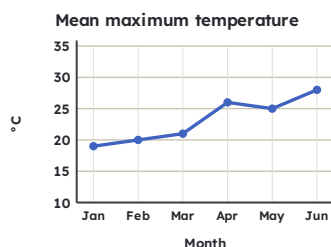
Medium

Questions 10–14

- 10 How many students were surveyed altogether?



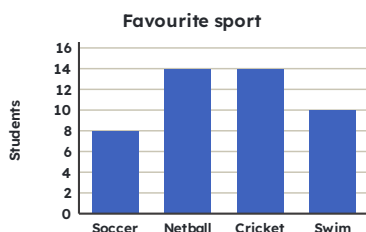
- 11 In which month was it warmest? By how much did it change from January to June?



- 12 Write 1 out of 4 as a percentage.

from 4.12

- 13 How many more students chose Cricket than Soccer?



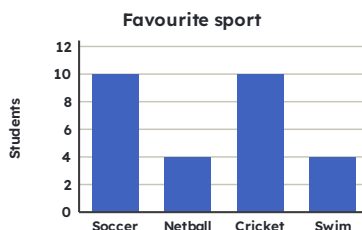
- 14 Tom got 23 out of 25 on a test. What percentage is that?

from 4.12

Challenging

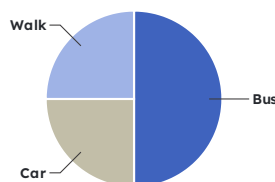
Questions 15–19

- 15 What percentage of the students chose Cricket? Round to a whole number.



- 16 80 students were surveyed. How many walk to school?

How students get to school

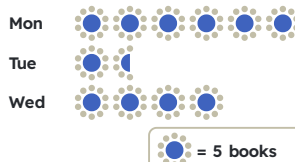


- 17 Write 250 mL as a fraction of 1 L.

from 4.12

- 18 Each symbol stands for 5 books. How many books were borrowed on Wed?

Library books borrowed



- 19 Where does $y = 4x - 5$ cross the y -axis?

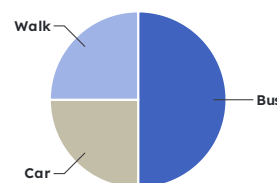
from 9.06

Challenging

Questions 20–24

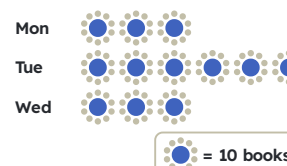
- 20 120 students were surveyed. How many walk to school?

How students get to school



- 21 Each symbol stands for 10 books. How many books were borrowed on Tue?

Library books borrowed



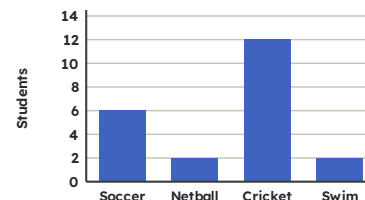
- 22 Which rule fits the table:
 $y = 4x + 3$, or $y = 3x + 4$, or $y = 5x + 2$?

x	1	2	3
y	7	11	15

from 9.05

- 23 What percentage of the students chose Soccer? Round to a whole number.

Favourite sport



- 24 Write 1 out of 25 as a percentage.

from 4.12

NAME _____ CLASS _____ DATE _____

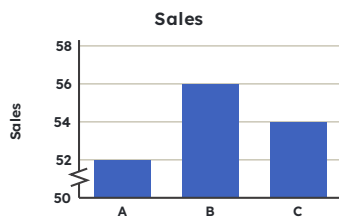
WE DO

Watch your teacher.
Copy every step.

KEY IDEA

A graph can mislead: a scale that does not start at 0, an uneven scale, a picture that grows in two directions, or missing labels. Always read the numbers, not just the size of the bars.

E1 Why is this graph misleading?



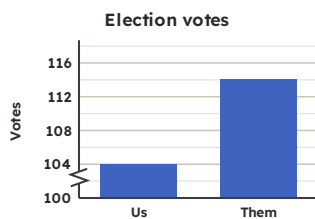
E2 Bar B looks 3 times as tall as bar A. Is B really 3 times A?

E3 A scale goes 0, 10, 20, 50, 100 with equal gaps. What is wrong?

Easy

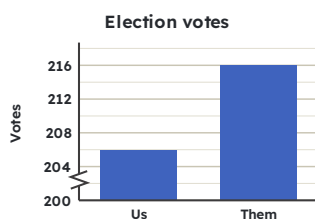
Questions 1–3

1 Where does the vertical scale start? Why could this mislead?



2 A picture graph uses a bigger picture for a bigger number, making it taller and wider. Why can this mislead?

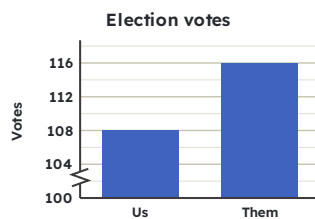
3 Where does the vertical scale start? Why could this mislead?



Easy

Questions 4–6

4 Where does the vertical scale start? Why could this mislead?

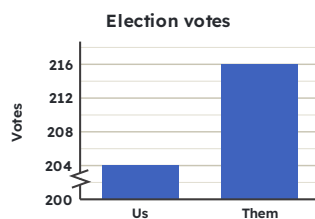


5 Bar A is 84 and bar B is 96. The scale starts at 80. How many times as tall does B look? How many times as big is B really?

looks: $\frac{96}{80} = 1.2$ times

really: $\frac{96}{84} = 1.14$ times

6 Where does the vertical scale start? Why could this mislead?



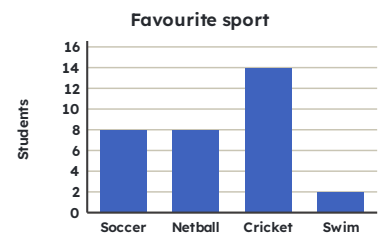
Medium

Questions 7–10

7 Bar A is 86 and bar B is 92. The scale starts at 80. How many times as tall does B look? How many times as big is B really?

8 A scale is marked 0, 100, 200, 500, 1000 with equal gaps. What is wrong with it?

9 How many more students chose Cricket than Netball?



from 10.01

10 Bar A is 44 and bar B is 52. The scale starts at 40. How many times as tall does B look? How many times as big is B really?

Medium

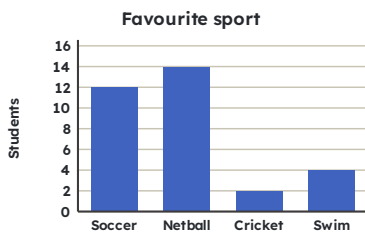
Questions 11–14

- 11 Bar A is 84 and bar B is 98. The scale starts at 80. How many times as tall does B look? How many times as big is B really?

- 12 Redraw this graph so it is not misleading. What would you change?



- 13 Which sport was the most popular?



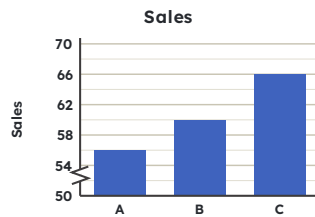
from 10.01

- 14 Bar A is 84 and bar B is 92. The scale starts at 80. How many times as tall does B look? How many times as big is B really?

Challenging

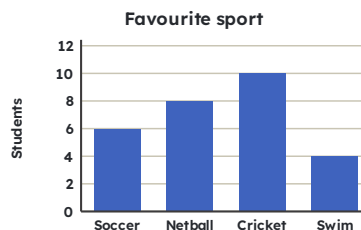
Questions 15–19

- 15 Redraw this graph so it is not misleading. What would you change?



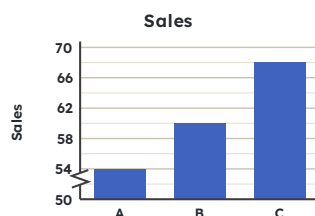
- 16 Ali's ad says sales "rocketed" from 28 to 31 thousand, with a graph whose scale starts at 27 thousand. Is "rocketed" fair? Explain.

- 17 How many students were surveyed altogether?



from 10.01

- 18 Redraw this graph so it is not misleading. What would you change?



- 19 In the table, what is y when $x = 1$?

x	1	2	3	4
y	5	7	9	11

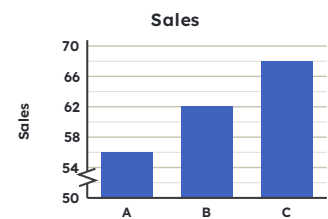
from 9.04

Challenging

Questions 20–24

- 20 Ali's ad says sales "rocketed" from 28 to 29 thousand, with a graph whose scale starts at 27 thousand. Is "rocketed" fair? Explain.

- 21 Redraw this graph so it is not misleading. What would you change?

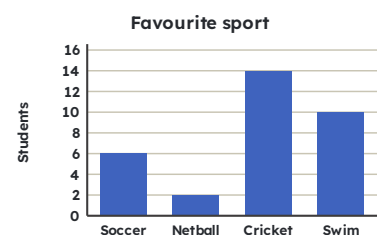


- 22 Start at D3. Move 1 left and 1 down. What is the grid reference now?

from 9.01

- 23 Tom's ad says sales "rocketed" from 22 to 25 thousand, with a graph whose scale starts at 21 thousand. Is "rocketed" fair? Explain.

- 24 Which sport was the most popular?



from 10.01

NAME _____ CLASS _____ DATE _____

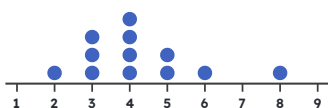
WE DO

KEY IDEA

In a dot plot each dot is one value, stacked above the number line. Count the dots for the number of values. The tallest stack is the most common value.

Watch your teacher.
Copy every step.

E1 How many values are in the dot plot?



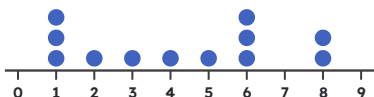
E2 Which value is the most common?

E3 Draw a dot plot of 1, 3, 3, 4, 4, 4, 6.

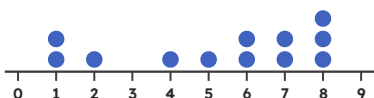
Easy

Questions 1–4

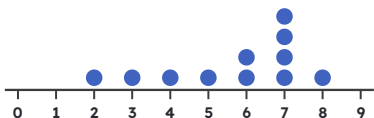
1 How many values are in the dot plot?



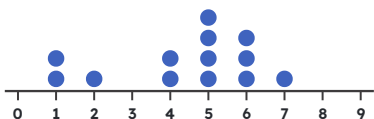
2 Which value is the most common?



3 How many values are more than 2?



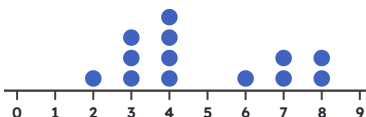
4 How many values are in the dot plot?



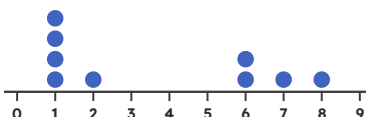
Easy

Questions 5–8

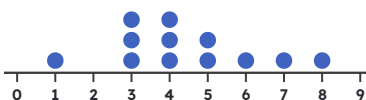
5 Which value is the most common?



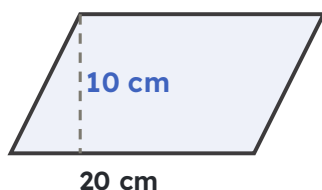
6 How many values are more than 7?



7 How many values are in the dot plot?



8 Find the area. (Not to scale.)

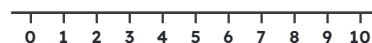


from 8.07

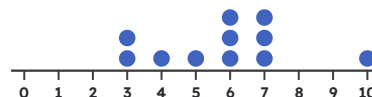
Medium

Questions 9–11

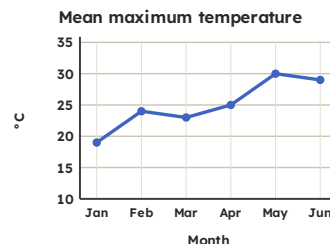
9 Draw a dot plot of 5, 8, 7, 7, 9, 4, 6, 9.



10 Is there an outlier (a value far from the others)? What is it?



11 In which month was it warmest? By how much did it change from January to June?

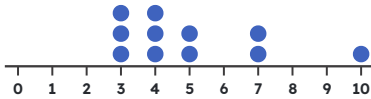


from 10.01

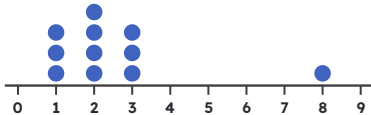
Medium

Questions 12–16

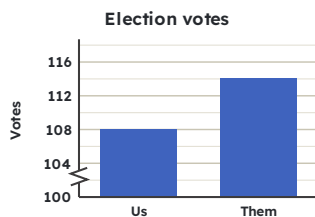
- 12 Is there an outlier (a value far from the others)? What is it?



- 13 What fraction of the values are less than 5?

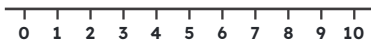


- 14 Where does the vertical scale start? Why could this mislead?

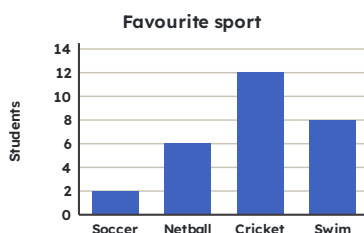


from 10.02

- 15 Draw a dot plot of 3, 5, 8, 9, 2, 4, 5, 5.



- 16 How many more students chose Cricket than Swim?

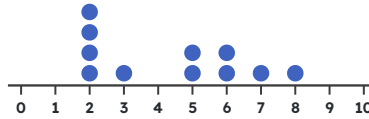


from 10.01

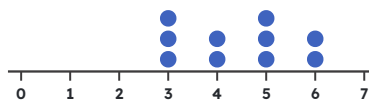
Challenging

Questions 17–21

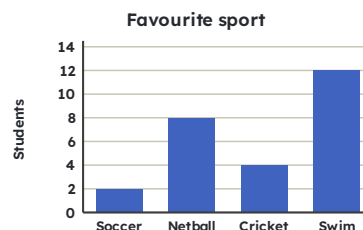
- 17 Describe the data: where is the cluster, and what is the spread?



- 18 Some students were asked how many pets they have. What percentage have 3 or more?

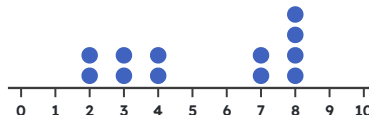


- 19 How many students were surveyed altogether?



from 10.01

- 20 Describe the data: where is the cluster, and what is the spread?



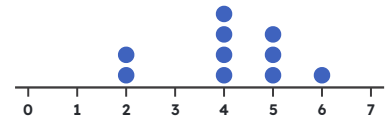
- 21 A 14 cm × 5 cm card has a 3 cm × 2 cm hole cut out. Find the area left.

from 8.08

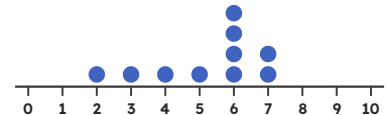
Challenging

Questions 22–26

- 22 Some students were asked how many pets they have. What percentage have 3 or more?



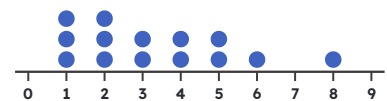
- 23 Describe the data: where is the cluster, and what is the spread?



- 24 Move (1, 4) 1 right, 6 up. Where is it now?

from 9.03

- 25 What fraction of the values are less than 6?



- 26 A picture graph uses a bigger picture for a bigger number, making it taller and wider. Why can this mislead?

from 10.02

NAME _____ CLASS _____ DATE _____

WE DO
KEY IDEA

In a stem-and-leaf plot, the stem is the tens digit and each leaf is a units digit: 3 | 5 means 35. An ordered plot has its leaves from smallest to largest.

Watch your teacher.
Copy every step.

E1 Write the values in the second row.

Stem	Leaf
2	4 8
3	2 5 7
4	1 6

Key: 2 | 4 = 24

E2 How many values are in the plot?

E3 Draw an ordered stem-and-leaf plot of 43, 27, 35, 41, 29, 38.

Easy

Questions 1–3

1 Write the values in the row with stem 4.

Stem	Leaf
1	3 6 8
2	4 4 7
3	8
4	2 3

Key: 1 | 5 = 15

2 How many values are in the plot?

Stem	Leaf
1	3 5 9
2	5
3	1 3 7
4	4

Key: 2 | 5 = 25

3 What are the smallest and the largest values?

Stem	Leaf
2	6
3	0 6 6 7
4	1 1 5
5	5

Key: 2 | 5 = 25

Easy

Questions 4–6

4 How many values are in the plot?

Stem	Leaf
1	5 7
2	5 6
3	0 5
4	4 8 9

Key: 2 | 5 = 25

5 What are the smallest and the largest values?

Stem	Leaf
1	3 5 7
2	
3	
4	0 2 4
5	0 2

Key: 2 | 5 = 25

6 Write the values in the row with stem 3.

Stem	Leaf
1	3
2	3 3 8
3	2 8 9
4	4 5 5 9

Key: 1 | 5 = 15

Medium

Questions 7–9

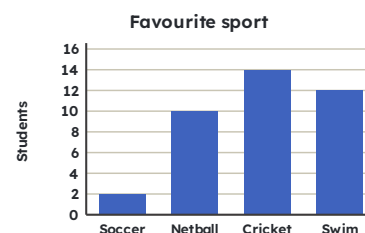
7 Draw an ordered stem-and-leaf plot of 39, 27, 28, 50, 22, 32, 34, 56.

8 How many values are 53 or more?

Stem	Leaf
3	0 2
4	0 0 5
5	5
6	0 4
7	5

Key: 3 | 5 = 35

9 How many students were surveyed altogether?



from 10.01

Medium

Questions 10–14

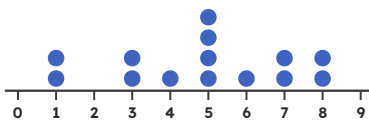
- 10 How many values are 62 or more?

Stem	Leaf
3	0 6
4	7
5	0 1 2 3 3
6	3 6
7	8

Key: 3 | 5 = 35

- 11 Draw an ordered stem-and-leaf plot of 37, 27, 31, 32, 48, 38, 54, 24.

- 12 How many values are more than 6?



from 10.03

- 13 These are test marks out of 100. The pass mark is 50. What percentage passed?

Stem	Leaf
4	1 5 9
5	6 8
6	0 4
7	
8	3 6 7

Key: 6 | 3 = 63

- 14 Bar A is 62 and bar B is 74. The scale starts at 60. How many times as tall does B look? How many times as big is B really?

from 10.02

Challenging

Questions 15–19

- 15 These are test marks out of 100. The pass mark is 50. What percentage passed?

Stem	Leaf
4	7 9
5	5 6 8
6	6 8
7	2 7
8	3

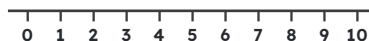
Key: 6 | 3 = 63

- 16 Which stem has the most leaves? What does that tell you?

Stem	Leaf
1	2 4 9
2	1 3 5 7
3	2 5

Key: 1 | 2 = 12

- 17 Draw a dot plot of 9, 8, 3, 4, 3, 2, 3, 8. from 10.03



- 18 These are test marks out of 100. The pass mark is 50. What percentage passed?

Stem	Leaf
5	9
6	1 1 7 8
7	0 0 7
8	6 8

Key: 6 | 3 = 63

- 19 Complete the table for $y = 3x + 3$.

x	0	1	2
y			

from 9.05

Challenging

Questions 20–23

- 20 Which stem has the most leaves? What does that tell you?

Stem	Leaf
1	9 9
2	7
3	0 1 3 4 8
4	6

Key: 1 | 2 = 12

- 21 These are test marks out of 100. The pass mark is 50. What percentage passed?

Stem	Leaf
4	4 6
5	7
6	2 2
7	1
8	2
9	0 0 3

Key: 6 | 3 = 63

- 22 Does (1, -1) lie on the line $y = 2x - 3$? Show the check. from 9.06

- 23 How many values are 54 or more?

Stem	Leaf
3	4
4	3 8 9
5	4 4 9
6	0
7	4 5 7

Key: 3 | 5 = 35

NAME _____ CLASS _____ DATE _____

WE DOWatch your teacher.
Copy every step.**KEY IDEA**

The mean is the total divided by how many values: the "fair share". The mode is the value that occurs most often (there can be more than one, or none).

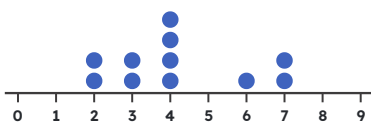
E1 Find the mean of 4, 7, 9, 6, 9.

E2 Find the mode of 4, 7, 9, 6, 9.

E3 The mean of 5 numbers is 8. Four of them are 6, 9, 7, 10. Find the fifth.

Easy

Questions 1–9

1 Find the mean of 12, 12, 9, 4, 8.**2** Find the mode of 7, 6, 4, 5, 2, 4, 1.**3** Find the total of 14, 14, 40, 24, then the mean.**4** Find the mean of 10, 10, 12, 6, 9, 1.**5** Find the mode of 3, 6, 6, 1, 9, 4, 6.**6** Find the total of 27, 15, 28, 30, then the mean.**7** Find the mean of 10, 6, 2, 6, 8, 4.**8** Find the mode of 3, 4, 4, 4, 6, 7, 1.**9** Which value is the most common?

from 10.03

Easy

Questions 10–18

10 Find the mode of 2, 8, 4, 1, 6, 3, 2.**11** Find the total of 32, 25, 25, 38, then the mean.**12** Find the mean of 8, 2, 7, 7.**13** A circle has a radius of 13 m. What is its diameter?

from 8.03

14 Find the mean of 23, 42, 12, 57, 50, 32.

total: ____

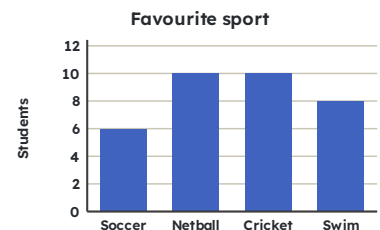
____ ÷ ____ = ____

15 Find the mode of 4, 5, 5, 7, 2, 3, 1.**16** Find the total of 28, 37, 10, 29, then the mean.**17** $79\,000 \div 1\,000$

from 3.03

18 Find the mean of 12, 6, 2, 9, 11.**Medium**

Questions 19–23

19 Find the mean of 18, 18, 43, 19, 33, 25.**20** Find the mean of 35, 20, 21, 29, 16, 40, to 1 decimal place.**21** How many students were surveyed altogether?

from 10.01

22 Mia scored 2, 9, 8, 8, 3 goals in five games. Find the mean and the mode.**23** Find the mean of 57, 41, 19, 27, 34, 12, 34.

Medium

Questions 24–32

- 24 Find the mean of 36, 23, 27, 38, 27, 21, to 1 decimal place.
-
-
- 25 Noah scored 10, 2, 2, 7, 9 goals in five games. Find the mean and the mode.
-
-
- 26 Change 280 000 m² to ha.
- from 8.04
- 27 Find the mean of 43, 42, 46, 45, 60, 45, 27.
-
-
- 28 What percentage of the class chose Fish?
- Favourite pet
-
- Dog
Cat
Bird
Fish
- from 10.01
- 29 The mean of five numbers is 10. Four of them are 12, 13, 7, 13. Find the fifth.
-
-
- 30 Find the mean of 21, 40, 22, 15, 31, 25, to 1 decimal place.
-
-
- 31 A graph has no labels on its axes. Why is that a problem?
- from 10.02
- 32 Ali scored 9, 3, 3, 10, 10 goals in five games. Find the mean and the mode.
-
-

Challenging

Questions 33–40

- 33 The mean of five numbers is 11. Four of them are 11, 7, 14, 11. Find the fifth.
-
-
- 34 Test marks: 77, 75, 72, 68. What must the next mark be to make the mean 75?
-
-
- 35 Does (2, 12) lie on the line $y = 4x + 4$? Show the check.
- from 9.06
- 36 Find the mean of 9, 6, 7, 9, 9. Then add 10 to every value. What is the new mean?
-
-
- 37 How many 7s are in 28?
- from 3.03
- 38 The mean of five numbers is 15. Four of them are 13, 16, 17, 16. Find the fifth.
-
-
- 39 Test marks: 79, 86, 71, 88. What must the next mark be to make the mean 83?
-
-
- 40 How many more students chose Soccer than Swim?
- Favourite sport
-
- Soccer Netball Cricket Swim
- from 10.01

Challenging

Questions 41–47

- 41 Test marks: 87, 85, 68, 84. What must the next mark be to make the mean 83?
-
-
- 42 Find the mean of 7, 10, 10, 6, 7. Then add 10 to every value. What is the new mean?
-
-
- 43 Round 24.8 to the nearest whole number.
- from 7.10
- 44 The mean of five numbers is 8. Four of them are 11, 11, 9, 5. Find the fifth.
-
-
- 45 Which sport was the most popular?
- Favourite sport
-
- Soccer Netball Cricket Swim
- from 10.01
- 46 Chloe scored 7, 8, 8, 10, 2 goals in five games. Find the mean and the mode.
-
-
- 47 Test marks: 80, 65, 70, 89. What must the next mark be to make the mean 78?
-
-

NAME _____ CLASS _____ DATE _____

WE DO

Watch your teacher.
Copy every step.

KEY IDEA

The median is the middle value when the data is in order: for an even number of values, it is halfway between the two middle ones. The range is the largest value minus the smallest.

E1 Find the median of 8, 3, 9, 5, 6.

E2 Find the median of 4, 10, 7, 2.

E3 Find the range of 12, 25, 18, 9, 21.

Easy

Questions 1–9

1 Find the median of 2, 12, 6, 16, 10, 19, 3.

2 Find the range of 5, 1, 4, 5, 8, 12.

3 Write 22, 14, 12, 24, 24, 12, 1 in order, smallest first.

4 Find the median of 13, 12, 19, 19, 5, 20, 12.

5 Find the range of 19, 30, 18, 7, 33, 43.

6 Write 28, 24, 8, 1, 28, 2 in order, smallest first.

7 Find the median of 4, 7, 6, 5, 2, 17, 14.

8 Find the range of 15, 24, 48, 45, 32, 39.

9 Find the mode of 7, 9, 5, 7, 2, 7, 3.

from 10.05

Easy

Questions 10–17

10 Find the range of 18, 40, 14, 23, 15, 18, 18.

11 Write 19, 25, 14, 10, 23, 21 in order, smallest first.

12 Find the median of 14, 11, 19, 4, 19.

13 Round 17.1 to the nearest whole number.

from 7.10

14 Find the median of 3, 29, 29, 11, 5, 23.

in order: _____

median: ____

15 Find the range of 30, 42, 14, 45, 15, 21, 13.

16 Write 13, 10, 23, 23, 10, 22, 5 in order, smallest first.

17 How many values are in the plot?

Stem	Leaf
1	4 8
2	1 2 2
3	6 9
4	1 4 8

Key: 2 | 5 = 25

from 10.04

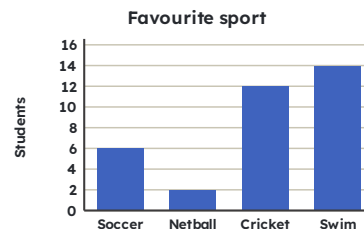
Medium

Questions 18–22

18 Find the median of 4, 27, 3, 29, 20, 9, 27, 23.

19 Find the median and the range of 42, 17, 53, 11, 34, 52, 54.

20 How many students were surveyed altogether?



from 10.01

21 Morning temperatures ($^{\circ}\text{C}$): $-3, 1, 3, -2, -6, 0$. Find the range.

22 Find the median of 26, 26, 11, 16, 12, 25, 17, 28.

Medium

Questions 23–32

- 23 Find the median and the range of 48, 14, 36, 29, 34.

- 24 Morning temperatures ($^{\circ}\text{C}$): 0, 3, -7, 4, -1, -5. Find the range.

- 25 How many millilitres does 102 cm^3 hold?

from 8.10

- 26 Find the median of 26, 13, 16, 27, 29, 10.

- 27 A graph has no labels on its axes. Why is that a problem?

from 10.02

- 28 The range of a set is 10. The largest value is 24. What is the smallest?

- 29 Find the median and the range of 12, 55, 19, 32, 34, 44, 17.

- 30 Find the total of 15, 25, 18, 18, then the mean.

from 10.05

- 31 Morning temperatures ($^{\circ}\text{C}$): 2, 0, -2, -3, -1, -2. Find the range.

- 32 Draw an ordered stem-and-leaf plot of 21, 43, 40, 36, 39, 38, 47, 43.

from 10.04

Challenging

Questions 33–39

- 33 The range of a set is 19. The largest value is 29. What is the smallest?

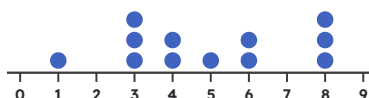
- 34 Find the mean and the median of 7, 7, 17, 19, 13, 7, 78. Which is the better "typical" value? Why?

- 35 A parallelogram has an area of 42 m^2 and a height of 7 m. Find its base.

from 8.07

- 36 Four numbers have a median of 11. The two middle numbers are 8 and $\square$. Find $\square$.

- 37 How many values are more than 4?



from 10.03

- 38 The range of a set is 7. The largest value is 19. What is the smallest?

- 39 Find the mean and the median of 25, 22, 22, 11, 12, 8, 93. Which is the better "typical" value? Why?

Challenging

Questions 40–46

- 40 Find the mean and the median of 16, 18, 10, 17, 25, 19, 78. Which is the better "typical" value? Why?

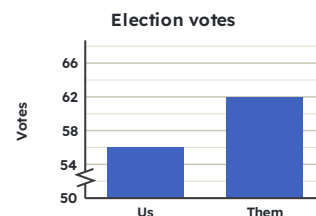
- 41 Four numbers have a median of 9.5. The two middle numbers are 8 and $\square$. Find $\square$.

- 42 Change 600 mm^2 to cm^2 .

from 8.04

- 43 The range of a set is 17. The largest value is 27. What is the smallest?

- 44 Where does the vertical scale start? Why could this mislead?



from 10.02

- 45 Morning temperatures ($^{\circ}\text{C}$): 2, -4, 3, -8, -6, 0. Find the range.

- 46 Find the mean and the median of 9, 10, 11, 18, 10, 15, 87. Which is the better "typical" value? Why?

NAME _____ CLASS _____ DATE _____

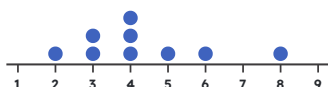
WE DO

KEY IDEA

To find statistics from a plot, first list the values in order (a dot plot or an ordered stem-and-leaf plot already shows them in order). Then find the range, median, mode and mean.

Watch your teacher.
Copy every step.

E1 Find the range and the mode.



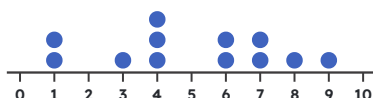
E2 Find the median.

E3 Find the mean.

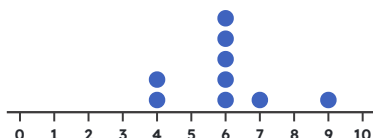
Easy

Questions 1–4

1 Find the range.



2 Find the mode.

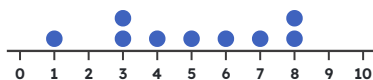


3 Find the range.

Stem	Leaf
3	1 2 4 4
4	3 9
5	1 2

Key: 2 | 4 = 24

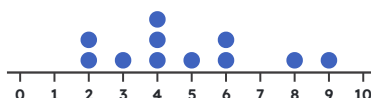
4 Find the range.



Easy

Questions 5–8

5 Find the mode.

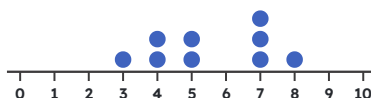


6 Find the range.

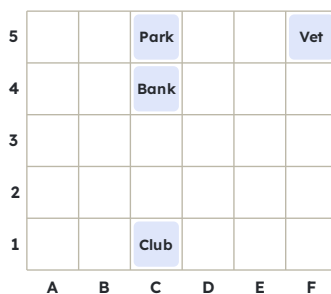
Stem	Leaf
2	3 5
3	0 8 8
4	6 8
5	0

Key: 2 | 4 = 24

7 Find the range.



8 Give the grid reference of the Bank.



from 9.01

Medium

Questions 9–11

9 Find the median.

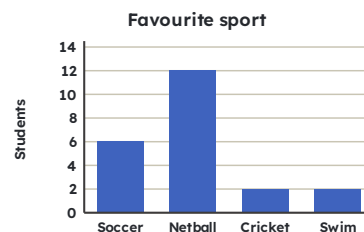


10 Find the median.

Stem	Leaf
2	1 3 7
3	
4	4 7
5	3 9

Key: 2 | 4 = 24

11 How many students were surveyed altogether?



from 10.01

Medium

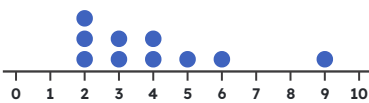
Questions 12–17

- 12 Find the median.

Stem	Leaf
2	4 9
3	1 7 8 9
4	7
5	4 4

Key: 2 | 4 = 24

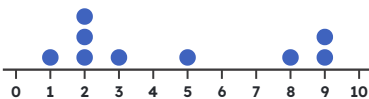
- 13 Find the mean.



- 14 Find the area of the triangular end: base 6 cm, height 4 cm.

from 8.12

- 15 Find the median.



- 16 A picture graph uses a bigger picture for a bigger number, making it taller and wider. Why can this mislead?

from 10.02

- 17 Find the mean, median and range.

Stem	Leaf
2	9
3	3 6
4	0 3 6
5	3 6

Key: 2 | 4 = 24

Challenging

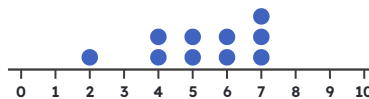
Questions 18–22

- 18 Find the mean, median and range.

Stem	Leaf
2	9
3	1
4	0 1
5	3 3 4 9

Key: 2 | 4 = 24

- 19 One more value of 10 is added. Which statistics change: range, median or mode?



- 20 Round \$3.121 to the nearest cent.

from 7.10

- 21 Find the mean, median and range.

Stem	Leaf
2	3
3	4 6 9
4	2 3 5
5	8

Key: 2 | 4 = 24

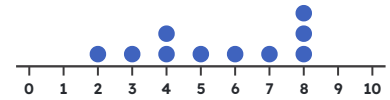
- 22 Find the total of 10, 34, 19, 29, then the mean.

from 10.05

Challenging

Questions 23–27

- 23 One more value of 10 is added. Which statistics change: range, median or mode?



- 24 Find the mean, median and range.

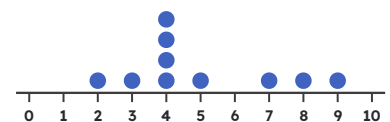
Stem	Leaf
2	6 9
3	1 9
4	2 5 6 6

Key: 2 | 4 = 24

- 25 Change 462 mL to litres.

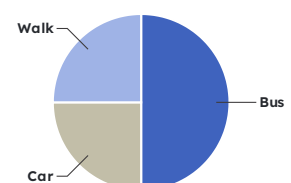
from 8.13

- 26 Find the mean.



- 27 What percentage of the class chose Bus?

How students get to school



from 10.01

NAME _____ CLASS _____ DATE _____

WE DOWatch your teacher.
Copy every step.**KEY IDEA**

To compare two sets of data, compare a centre (the mean or median: which is higher?) and the spread (the range: which is more consistent?). Then write a sentence about what it means.

E1 Class A: 5, 7, 8, 8, 9. Class B: 3, 6, 8, 9, 10. Compare the medians.

_____**E2** Compare the ranges. Which class is more consistent?

_____**E3** Which class did better overall? Use the mean.

_____**Easy**

Questions 1–8

1 Set A: 6, 14, 4, 20, 19. Set B: 13, 6, 15, 11, 13. Which set has the larger range?
_____**2** Set A: 5, 9, 12, 10, 6. Set B: 11, 3, 6, 7, 8. Which set has the higher median?
_____**3** Set A: 12, 20, 20, 2, 14. Set B: 15, 7, 5, 17, 7. Which set has the larger range?
_____**4** Set A: 15, 14, 17, 1, 3. Set B: 18, 3, 19, 6, 7. Which set has the higher median?
_____**5** Set A: 10, 3, 14, 15, 3. Set B: 19, 8, 3, 9, 9. Which set has the larger range?
_____**6** Set A: 16, 2, 10, 12, 19. Set B: 13, 16, 4, 9, 20. Which set has the higher median?
_____**7** Set A: 5, 8, 15, 10, 18. Set B: 3, 14, 13, 15, 3. Which set has the larger range?
_____**8** Set A: 15, 12, 20, 1, 2. Set B: 7, 1, 8, 12, 16. Which set has the higher median?
_____**Easy**

Questions 9–16

9 Set A: 7, 20, 3, 13, 1. Set B: 12, 10, 19, 19, 10. Which set has the higher median?
_____**10** Set A: 19, 20, 11, 12, 10. Set B: 8, 9, 6, 10, 2. Which set has the larger range?
_____**11** Team A scored 48, 76, 56, 66, 49. Team B scored 73, 75, 51, 43, 73. Compare the means.
A: ____, B: ____
_____**12** Round 68.3 to the nearest whole number.

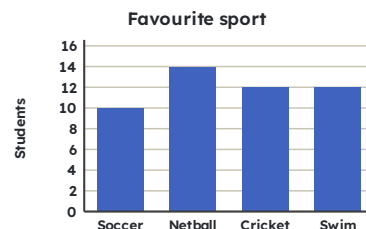
from 7.10

13 Set A: 6, 11, 8, 7, 15. Set B: 13, 3, 18, 4, 19. Which set has the higher median?
_____**14** Set A: 17, 5, 13, 18, 10. Set B: 13, 20, 16, 8, 1. Which set has the larger range?
_____**15** Team A scored 63, 65, 61, 47, 44. Team B scored 43, 65, 56, 69, 72. Compare the means.
_____**16** Find the range of 23, 39, 48, 30, 32, 41.

from 10.06

Medium

Questions 17–21

17 Team A scored 49, 65, 50, 45, 71. Team B scored 48, 72, 56, 68, 41. Compare the means.
_____**18** Runner A's times (s): 13, 14, 14, 15, 15. Runner B's: 8, 12, 14, 17, 19. Who is more consistent? Explain.
_____**19** How many students were surveyed altogether?
_____

from 10.01

20 Team A scored 75, 53, 72, 70, 55. Team B scored 54, 73, 71, 50, 62. Compare the means.
_____**21** Runner A's times (s): 12, 13, 13, 14, 14. Runner B's: 7, 11, 13, 16, 18. Who is more consistent? Explain.

Medium

Questions 22–28

- 22 Runner A's times (s): 9, 10, 10, 11, 11. Runner B's: 4, 8, 10, 13, 15. Who is more consistent? Explain.

- 23 Team A scored 80, 66, 50, 64, 70. Team B scored 78, 50, 45, 41, 41. Compare the means.

- 24 Complete the table for $y = x + 5$.

x	0	1	2
y			

from 9.05

- 25 Compare the two classes using the medians and the ranges.

Class A	Stem	Class B
6 1	2	
6	3	5 6 8
4 1 0 0	4	5 6
	5	5 6

Key: 2 | 4 = 24

- 26 A picture graph uses a bigger picture for a bigger number, making it taller and wider. Why can this mislead?

from 10.02

- 27 Runner A's times (s): 11, 12, 12, 13, 13. Runner B's: 6, 10, 12, 15, 17. Who is more consistent? Explain.

- 28 Team A scored 41, 47, 74, 45, 68. Team B scored 80, 60, 61, 42, 62. Compare the means.

Challenging

Questions 29–34

- 29 Compare the two classes using the medians and the ranges.

Class A	Stem	Class B
1	2	7 8
9 7 4	3	0 4 4
7 5 2	4	4
	5	9

Key: 2 | 4 = 24

- 30 Two dice were each rolled 6 times. Die A: 6, 4, 2, 4, 1, 1. Die B: 2, 5, 2, 5, 1, 3. Compare the mean and range of each.

- 31 Check Euler's rule $F + V - E = 2$ for a triangular prism.

from 8.09

- 32 Compare the two classes using the medians and the ranges.

Class A	Stem	Class B
8 1 0	3	5 5 6 8
7 5 1 1	4	3
	5	7 9

Key: 2 | 4 = 24

- 33 Find the total of 10, 11, 30, 17, then the mean.

from 10.05

- 34 Two dice were each rolled 6 times. Die A: 3, 3, 6, 1, 5, 6. Die B: 6, 6, 4, 2, 1, 5. Compare the mean and range of each.

Challenging

Questions 35–40

- 35 Two dice were each rolled 6 times. Die A: 6, 5, 5, 4, 3, 1. Die B: 4, 4, 5, 6, 4, 1. Compare the mean and range of each.

- 36 Compare the two classes using the medians and the ranges.

Class A	Stem	Class B
7 0	2	
	3	4 5 8
9 6 5 4 1	4	1 5
	5	0 4

Key: 2 | 4 = 24

- 37 Change 2 621 mL to litres.

from 8.13

- 38 Two dice were each rolled 6 times. Die A: 3, 1, 2, 1, 3, 2. Die B: 2, 2, 6, 6, 3, 5. Compare the mean and range of each.

- 39 How many values are in the plot?

Stem	Leaf
2	0 0 2 3
3	1 8
4	1 5 6

Key: 2 | 5 = 25

from 10.04

- 40 Compare the two classes using the medians and the ranges.

Class A	Stem	Class B
7 6 6 3 2	2	5
4	3	3 3 9
8	4	2 7
	5	7

Key: 2 | 4 = 24